Teng Lei

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EDUCATION

Guangdong Technion-Israel Institute of Technology, Shantou, China Sep. 2023 - Present

Bachelor of Science in Mathematics with Computer Science *Expected in Jun. 2027*

- **GPA**: 96.9/100; Ranking: 1/74
- **Core Coursework**: Computer Architecture, Operating Systems, Digital Systems, Data Structures, Programming Languages, Combinatorial Algorithms, Mathematical Statistics, Introduction to Numerical Analysis
- **Honors**: Vice Chancellor's List Scholarship, Second Class Chancellor's Scholarship

PUBLICATIONS

1. Liu, Y., Zheng, Y., Ye, W., **Lei, T.**, Tang, H., & Mou, Z. (2026). An efficient lightweight Spike Neuron Network applied to Post-stroke Dysarthria speech recognition, *Speech Communication*, Volume 181, 2026, 103399. https://doi.org/10.1016/j.specom.2026.103399. (Accepted)
2. Li, J., Liang, Y., Ye, W., **Lei, T.**, Wang, P., Zhang, G., & Liu, Y. (2026). Dual-branch AIGC image detection leveraging integrated semantic and frequency-domain features, *Computer Vision and Image Understanding*, Volume 271, 2026, 104893. https://doi.org/10.1016/j.cviu.2026.104893. (Accepted)

RESEARCH EXPERIENCE

Dual-branch AIGC image detection leveraging integrated semantic and frequency-domain features 2026

Role: 5th Author

Validated the proposed AIGC image detection framework across diverse GAN and diffusion-generated datasets, assessing its cross-model generalization.

- Cleaned and preprocessed 33 image datasets from different generative models for experimental evaluation.
- Contributed to manuscript review and revision, refining technical descriptions and the presentation of experimental findings.

An efficient lightweight Spike Neuron Network applied to Post-stroke Dysarthria speech recognition 2026

Role: 4th Author

- Responsible for reproducing and evaluating baseline models (VGG11, ResNet18, Conformer, Spikformer) on the Post-stroke Dysarthria (PSD) dataset.
- Assisted in the data preprocessing pipeline, extracting 39-dimensional MFCCs (including delta and delta-delta coefficients) to replace compute-heavy spectrograms, reducing input FLOPs.
- Conducted ablation studies on temporal steps (T = 1, 2, 4) of LIF neurons to verify energy efficiency versus accuracy trade-offs.

Algebraic equations of low degree and introduction to Galois theory Oct. 2025 - Jun. 2026

Advisor: Boris Shapiro

- Advanced readings on the algebraic and topological aspects of polynomial solvability. Studied topics including solutions of cubic and quartic equations, Abel–Ruffini theorem, insolvability of the general quintic, monodromy groups, Riemann surfaces of algebraic functions, and Arnold's elementary and topological approaches to equation theory.

Automatic Recognition of Mathematical Blackboard Writing with LaTeX Output Nov. 2025 - Mar. 2026

NLP Project | Advisor: Prof. Antti Rasila

- Planned to design and implement a specialized language modeling framework for recognizing handwritten mathematical blackboard content, combining OCR preprocessing, BPE-based tokenization, and Transformer architectures to handle complex symbolic and structural patterns.
- Implemented an end-to-end recognition pipeline for handwritten mathematical blackboard notes, converting noisy visual inputs into structured LaTeX outputs to enhance clarity and interpretability.

PROJECT EXPERIENCE

Shanghai Jiao Tong University - AI4Math Summer School Aug. 2026

- Studied formal mathematics and Lean 4, learning to represent mathematical concepts and proofs in a computer-verifiable form and gaining understanding of mathematical logic and programming.
- Explored AI mathematical reasoning, including automated proof generation, proof verification, and the integration of large language models with formal mathematics systems.
- Contributed to **TRACER**, an AI-assisted mathematical proof repair agent, by developing a compiler feedback module, strengthening proof safety checks, and implementing automated testing to help identify and correct proof errors.

Operating System Design and Implementation - xv6 Kernel Projects Nov. 2025 - Dec. 2025

Class Project | Advisor: Prof. Guillaume Hoffmann

- Extended the xv6 operating system with modern OS features, including null pointer checking, memory address protection, and multithreaded concurrency support.
- Modified core kernel components, including process management and memory handling, and verified the functional completeness and correctness of the system.
- Achieved full marks for the project through rigorous testing and validation of system-level enhancements.

Implementing the Huarong Dao Puzzle Game in Z80 Assembly Dec. 2024 - Jan. 2025

Class Project | Advisor: Prof. Guillaume Hoffmann

- Led a team of four to develop the Huarong Dao puzzle game from scratch using Z80 assembly language, implementing core gameplay mechanics including block selection, release, collision detection, and movement.
- Designed 10 progressively challenging levels and implemented a step-counting and scoring system to enhance gameplay structure.
- Performed extensive debugging and low-level optimization to improve correctness and execution stability.

WORK EXPERIENCE

GAC R&D Center Guangzhou, China

Data Analyst Intern Sep. 2024 - Oct. 2024

- Processed and analyzed 100,000+ real-world vehicle driving records, performing data cleaning, classification, and statistical aggregation.
- Examined behavioral patterns under special conditions (e.g., low-battery trips) to support data modeling for intelligent vehicle systems development.
- Wrote and optimized SQL queries to extract and aggregate large-scale driving data from multiple databases.
- Delivered multiple data summary reports based on large-scale vehicle datasets, providing quantitative insights for data-driven decision-making.

LANGUAGES & SKILLS

Programming Languages: C/C++, Python, Java, SQL, Haskell; Assembly: Z80, x86, MIPS

Systems & Tools: Unix/Linux, MySQL, Git, Lean4 proof assistant

Machine Learning Frameworks: PyTorch, TensorFlow